

Mohammed Ameen Naimuddin

[GitHub](#)[LinkedIn](#)ameennaimuddin.dev@gmail.com[+91-9701755616](tel:+91-9701755616)

SUMMARY

Aspiring Machine Learning Engineer with hands-on experience in building end-to-end AI systems using transformer models, vector search, and deep learning. Skilled in Sentence-BERT, FAISS, FastAPI, and Streamlit, with a strong foundation in NLP, computer vision, and scalable ML system design.

EXPERIENCE

Rubixe

Sep 2025 – Apr 2026

- Built **end-to-end ML pipelines** for credit risk, healthcare analytics, and demand forecasting
- Developed a **credit scoring model** using Logistic Regression & XGBoost, achieving strong performance
- Developed solutions for **price prediction, healthcare risk analysis, insurance cost estimation, and demand forecasting**
- Performed **EDA, feature engineering, and data transformations**, improving model accuracy and robustness
- Optimized models via **feature importance, encoding, and hyperparameter tuning**, reducing misclassification
- Evaluated models using **AUC, Gini, and R^2 (0.88+)**, delivering actionable insights for data-driven decision making

NullClass

Jul 2024 – Sep 2024

- Built a **CNN-based emotion detection system** to classify facial expressions from image data
- Applied **image preprocessing and augmentation** to improve model generalization
- Trained deep learning models to recognize emotions such as **happy, sad, angry, and neutral**
- Evaluated performance using **accuracy and confusion matrix** and improved results through tuning

PROJECTS

Lung Cancer Detection using CNN

- Trained an Xception-based CNN achieving **93%+ accuracy** on lung cancer classification
- Applied **data augmentation, EarlyStopping, ReduceLROnPlateau** to improve generalization
- Built complete pipeline: preprocessing, training, evaluation, and visualization
- Enhanced feature extraction using **fuzzy logic preprocessing**

MediBot

[Live Demo](#)

- Developed AI-powered chatbot for **symptom analysis and preliminary medical assistance**
- Built modules for **appointment scheduling and diet recommendations**
- Designed user-friendly interface improving accessibility and usability
- Led feature development and system flow optimization

Semantic Search Engine

[Live Demo](#)

— [GitHub](#)

- Developed a **semantic search engine** using Sentence-BERT embeddings for context-aware retrieval
- Implemented **FAISS vector indexing** enabling fast similarity search on large text datasets
- Designed scalable architecture for semantic retrieval using vector embeddings and efficient indexing
- Designed REST APIs using **FastAPI** for efficient query processing and embedding retrieval
- Built an interactive UI using **Streamlit** for real-time query testing
- Improved search relevance and reduced query latency using **cosine similarity on dense vector embeddings**
- Structured modular pipeline: embedding → indexing → retrieval → UI

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **ML/AI:** BERT, Sentence-Transformers, CNN, FAISS, NLP, Deep Learning
- **Frameworks:** FastAPI, Streamlit, TensorFlow/Keras, Scikit-learn
- **Tools:** Git, VS Code
- **Concepts:** Semantic Search, Vector Databases, Embeddings, Model Evaluation, API Design

CERTIFICATIONS

- Career Essentials in Generative AI – Microsoft & LinkedIn
- Attended **AI Days 2024 – AI for Society** at **JNTU Hyderabad**, exploring real-world AI applications, emerging trends, and industry use cases through expert sessions and workshops.
- Programming with Python – Internshala (2023): Completed 6-week training covering OOP, SQLite, and GUI development; scored **90% (Top Performer)**

EDUCATION

2021 – 2025	BTech, Computer Science and Engineering (AIML) SRM-AP University
2019 – 2021	Class 12th State Board Of Intermediate Education Narayana Junior College
2018 – 2019	Class 10th Central Board of Secondary Education Sri Chaitanya School